

Spontaneous Symmetry Breaking and Degenerate Vacua from the 90-Degree Floor——Symmetry Breaking by an Infinitesimal Seed and the $\Delta = -2\pi/N$ Lock ($N = 6, \dots, 40$, all cases)

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Abstract

The 90-degree floor is a self-consistent unstable relative equilibrium with a 90° phase skeleton (Papers 1 and 2). When the same floor Z_0 is given an infinitesimal seed of amplitude 10^{-8} pointing in **different directions** and run with the exact one_step, the symmetry breaks, and after exponential inflation (Paper 6) the system locks onto an equal-amplitude, phase-dispersed, tilted relative equilibrium (one of the degenerate vacua). In all 35 cases $N = 6, \dots, 40$, every seed triggers inflation and locks onto an **isomorphic equilibrium** with a 1-step rotation angle that is exactly $\Delta = -2\pi/N$ (error $\sim 10^{-5}$) and equal amplitude (ratio $\rightarrow 1$, $\sim 10^{-3}$), while the phase configuration (the direction of the vacuum) differs depending on the seed (vacuum difference: median 4.0° – 48.4° , maximum 76.9° – 180°). That is, from a symmetric, unstable floor, an infinitesimal seed **spontaneously selects** one of the degenerate vacua. This is spontaneous symmetry breaking on a discrete map (Mexican-hat type). From the measured dynamics of the order parameter $r = \sqrt{H_\perp/H}$, a Landau-type effective potential $V(r)$ can be reconstructed with measured parameters ($N = 6: \lambda = 0.338, r_{\text{vac}} = 0.485$ / $N = 40: \lambda = 0.076, r_{\text{vac}} = 0.330$; ansatz). Large N that were unlocked at 500 steps all lock within 2000 steps ($N = 22$ at step 663, $N = 40$ at step 1299). Classification: empirical support for a new hypothesis (the SSB picture; exhaustive verification).

1. Problem

In Paper 1 the 90-degree floor was established as “an unstable relative equilibrium in which the amplitude-driving $\sin(2\Delta\varphi)$ vanishes individually for every term (90° phase skeleton)”, and in Paper 2 as “the qualification for ignition = an unstable relative equilibrium”. This paper investigates what that instability concretely produces——**what an infinitesimal seed selects**——for all N , and reads it as Mexican-hat-type SSB.

2. Dynamics and the seed (control experiment)

The map is the canonical one_step (Paper 1 §2, SHA 1abf2353...). The floor Z_0 is step0 of full_N3_N40_sweep (the make_parent eigenmode floor, Paper 2 §6). The only control variable is the initial seed (no change to the physical equations; the decision rules are not put into the dynamics but applied in post-run analysis): for each N we give $Z_0 + 10^{-8}(\xi + i\eta)$ (ξ, η random, normalized) in 5 different directions, and run until adaptive stopping (equal amplitude and gauge-invariant relative-phase rate of change below threshold, upper limit 2600 steps). In the final state of each seed we measure lock_step, $H_\perp/H$, amplitude ratio, Δ (1-step rotation angle), $|\Delta + 2\pi/N|$, and the inter-seed difference of the relative-phase configuration.

3. Mechanism——Mexican-hat-type SSB

The 90-degree floor = the 90° skeleton unstable relative equilibrium (schematically, the summit of the sombrero); the valley = the family of degenerate relative equilibria (equal amplitude, phase dispersion, tilted equilibria; sharing the common invariants $\Delta = -2\pi/N$ and equal amplitude). An infinitesimal seed breaks the symmetry and rolls the system down an unstable eigendirection (inflation, Paper 6), locking onto a single point in the valley——from an unstable symmetric maximum it spontaneously selects one of the degenerate lower equilibria. We confirmed for $N = 6$ that the valley (the degenerate vacuum manifold) is not a finite discrete map-symmetry orbit but a **continuous moduli of tilted relative equilibria** (the 5 final states do not map onto one another under any of the $6! = 720$ vertex permutations plus a global phase plus complex conjugation: 頂点置換による縮退真空同定_N6_20260913/). This is consistent with the fact that the valley of a Mexican-hat is intrinsically a continuous degenerate manifold (this is not the claim that the valley forms a single discrete symmetry orbit). From the measurement of the order parameter $r = \sqrt{H_\perp/H}$ ($H_\perp$ = the energy of the component orthogonal to the floor plane), a Landau-type

$$V(r) = -\frac{1}{2}\lambda r^2 + \frac{\lambda}{4r_{\text{vac}}^2}r^4, \quad \lambda = \ln|\mu_{\text{max}}|, \quad r_{\text{vac}} = \sqrt{H_\perp/H|_{\text{sat}}}$$

can be reconstructed ($N = 6:\lambda = 0.338, r_{\text{vac}} = 0.485$ / $N = 40:\lambda = 0.076, r_{\text{vac}} = 0.330$). This is an **effective description (ansatz)** in which the coefficients are identified from measured quantities. A first-principles derivation of $V(r)$ is left as future work. $r = \sqrt{H_\perp/H}$ is one scalar coordinate representing the amount of departure from the floor, and the Mexican-hat in the figures is a **schematic visualization** of the SSB configuration; it does not mean that r is an exact radial coordinate common to all degenerate vacua (in fact $H_\perp/H$ differs from vacuum to vacuum, 0.19–0.38, §7). Here “vacuum” is a mathematical

designation for the degenerate minima of the effective potential; the physical characterization of the vacuum is left as future work. Note that $\lambda = \ln |\mu_{\max}|$ ($\mu_{\max}$ = the spectral radius of the floor Jacobian, a quantity distinct from the μ of iK) is itself the inflation amplification rate of Paper 6, so **the curvature of the SSB potential = the amplification rate of inflation** are tied together by the same $\mu_{\max}$.

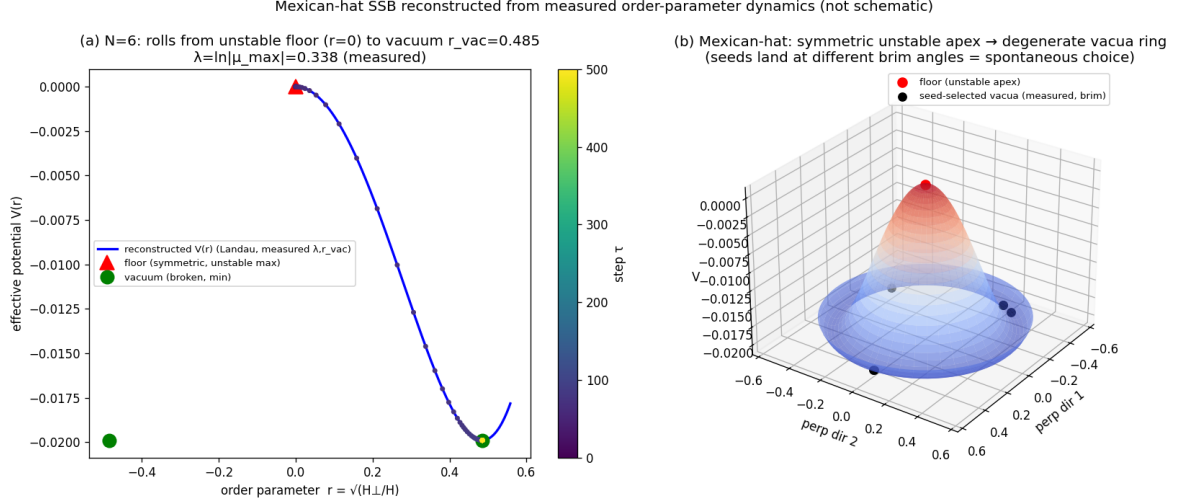


Figure 1: Mexican-hat-type SSB (measured reconstruction, $N=6$). Left = the measured order parameter falling on $V(r)$ from the floor (unstable summit) to the valley; right = the sombrero surface and the degenerate vacuum ring (5 seed points).

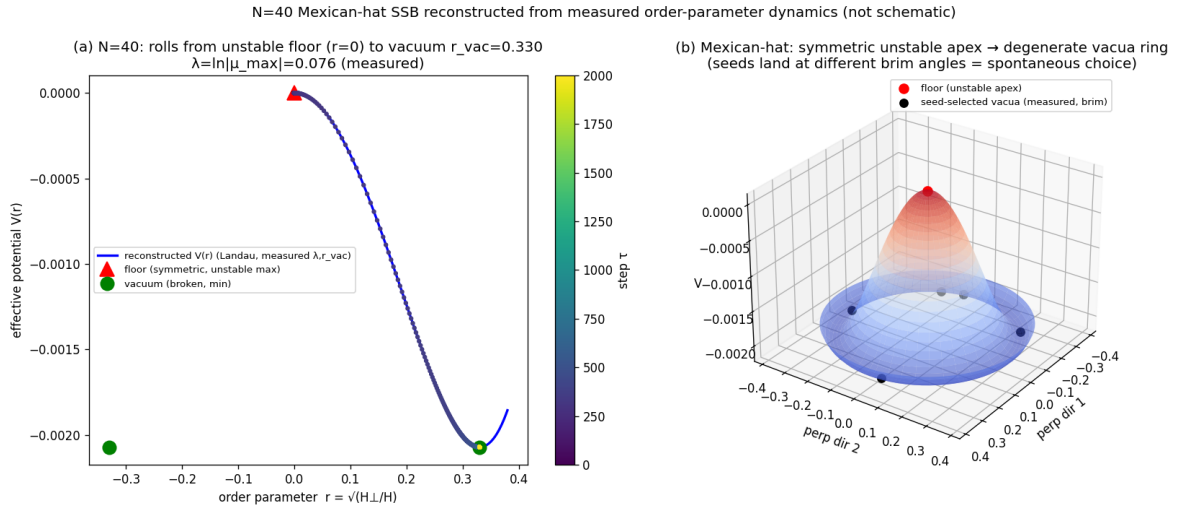


Figure 2: The same for $N=40$ ($\lambda=0.076$, $r_{\text{vac}}=0.330$).

Figures 1 and 2 Mexican-hat-type SSB (measured reconstruction, $N = 6 \text{ } / \text{ } N = 40$).

4. Decisive verification (same floor + different seeds $\rightarrow$ degenerate vacua, $N = 6$)

Final states obtained by giving the $N = 6$ floor seeds of amplitude 10^{-8} pointing in different directions: every seed locks onto an **isomorphic equilibrium** with $\Delta = -1.047198 = -2\pi/6$ ($|\Delta + 2\pi/N| \leq 4.3 \times 10^{-7}$) and amplitude ratio 1.0000–1.0006. Meanwhile, the inter-seed difference in the direction of the vacuum (the relative-phase configuration) **differs**, from median 1.6° (seed1 vs 0) to 112.7° (seed5 vs

0). The invariants ($\Delta = -2\pi/N$, equal amplitude) are common to the vacua, while the broken degree of freedom (the direction of the phase configuration) is selected by the seed.

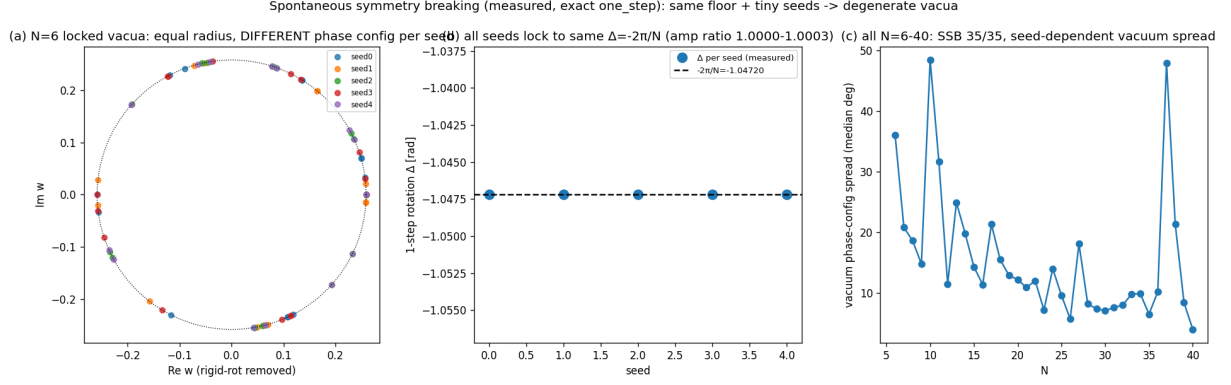


Figure 3: $N=6$, same floor + 5 different infinitesimal seeds. Left = the locked configuration of each seed (rigid rotation removed; equal radius but different phase configurations = degenerate vacua), center = all seeds $\Delta=-2\pi/N$, right = the N -dependence of the vacuum difference over all $N=6-40$.

Figure 3 SSB degenerate vacua (real data, $N = 6$).

5. Exhaustive verification ($N = 6, \dots, 40, 35/35$)

For all 35 cases $N = 6-40$ (5 differently-directed seeds per $N = 35 \times 5 = 175$ runs in total), all three SSB conditions hold (batch summary `ssb_batch_summary.csv`):

- **Every seed triggers inflation** (clear departure from the floor, `onset_ok=True` for all 35).
- **All equal amplitude** (amplitude ratio < 1.01 , `all_equal_amp=True` for all 35) and **the 1-step rotation angle is exactly $\Delta = -2\pi/N$** (`all_lock_2piN=True` for all 35, $|\Delta + 2\pi/N| < 10^{-3}$).
- **The phase configuration (the direction of the vacuum) differs depending on the seed:** vacuum difference median 4.0° ($N = 40$)– 48.4° ($N = 10$), maximum 76.9° ($N = 26$)– 180.0° .

That is, from an unstable floor, an infinitesimal seed spontaneously selects one of the degenerate vacua permitted by the symmetry of the map (SSB confirmed in 35/35).

Figure 4 SSB over all $N = 6-40$ (35/35).

6. Long-time verification ($N = 22, 40, 2000$ steps)

The large N ($N = 22, 30, \dots, 40$) that were unlocked at 500 steps were so because of truncation; running without stopping up to 2000 steps, they all lock. $N = 22$ locks completely at step 663 and $N = 40$ at step 1299 (Δ_{tail} exactly $-2\pi/N$, equal-amplitude ratio 1.0000, rigid-rotation residual of floor/tail at machine zero). A control test confirmed that the initial value bit-matches the old run at step0 (step1 difference $\sim 10^{-16}$, after which inflation exponentially amplifies), ensuring faithful reproduction.

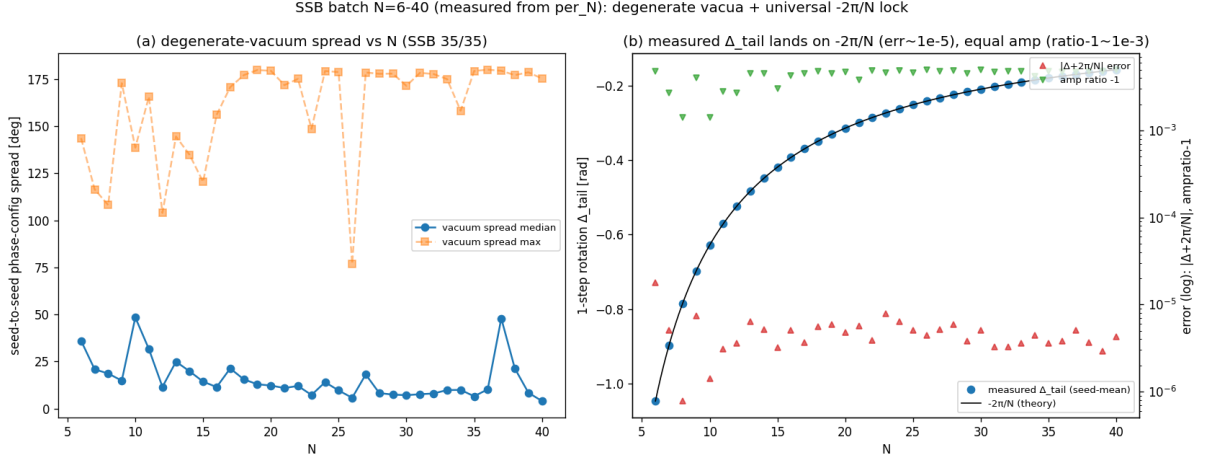


Figure 4: SSB batch over all N6-40. Left = the N-dependence of the vacuum difference (median/maximum); right = the measured Δ_{tail} agreeing with the theoretical $-2\pi/N$ for all N (error $\sim 1e-5$) and amplitude ratio $\rightarrow 1$.

7. Claim classification and open items

- Classification: empirical support for a new hypothesis (the SSB picture; exhaustive verification). Verdict: retained.
- Assertion: this system exhibits spontaneous symmetry breaking (the configuration—the 90° skeleton unstable relative equilibrium $\rightarrow$ the family of degenerate lower equilibria $\rightarrow$ spontaneous selection by an infinitesimal seed $\rightarrow$ all vacua isomorphic with $\Delta = -2\pi/N$ and equal amplitude—holds to machine precision, 35/35. The Mexican-hat is a schematic effective model, §3). **The key point is that from the same 90° floor, the terminal surface (the direction of the vacuum) is not uniquely determined but depends on the seed.**
- Structure of the vacuum manifold (the valley): the 5 final states of $N = 6$ do not map onto one another under any of the $6! = 720$ vertex permutations plus a global phase (plus complex conjugation) (minimum residual 0.05–0.83). Hence the valley is not a finite discrete map-symmetry orbit but a **continuous moduli of tilted relative equilibria** ($H_\perp/H = 0.19\text{--}0.38$ is continuous). This is consistent with the fact that the valley of a Mexican-hat is intrinsically a continuous degenerate manifold, and does not negate SSB. Identifying the dimension and metric of the moduli is left as future work.
- A detailed classification of the symmetry group is outside the scope of this paper. The invariants are $\Delta = -2\pi/N$ and equal amplitude; what changes with the seed is the direction of the vacuum (tilt $H_\perp/H$ is 0.19–0.38, $N = 6$). “Vacuum” is a mathematical designation for the degenerate minima of the effective potential; the physical vacuum is uncharacterized. $V(r)$ is an ansatz.

Related work (structural agreement, not the source of derivation)

- Spontaneous symmetry breaking / Landau theory / Goldstone • Higgs (Landau 1937; Goldstone 1961; Anderson 1963; Higgs 1964): this section is one example of SSB on a discrete map, and the reconstructed effective potential is a Ginzburg–Landau-type sombrero.

- Kuramoto model (synchronization): lock = synchronization.

References

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Reproduction

The verification scripts, figures, and summaries are in 位相ロック全 N 確認_相対平衡_20260912/ 自発的対称性の破れ_シード依存縮退真空_20260912/ (verify_ssb_seed_vacua_20260912.py, make_figure_ssb.py, make_figure_mexican_hat.py/_N40.py, REPORT.md, SHA256SUMS.txt, run_all.sh), and under it 全 $N6_40_SSB$ バッチ_20260912/ (run_ssb_batch_N6_N40_20260912.py, per_N/N{6..40}.txt, ssb_batch_summary.csv, fig_ssb_batch_allN.png). The long-time runs are N22_2000step 検証走行_20260912/ and N40_2000step 検証走行_20260912/. The decisive $N = 6$ test that the vacuum manifold is a continuous moduli (not a finite discrete symmetry orbit) is 頂点置換による縮退真空同定_N6_20260913/ (verify_vacua_permutation_orbit_N6_20260913.py = imports the canonical one_step with SHA verification, a read-only exhaustive S_6 sweep). The floor data is the make_parent floor (Paper 2 §6). The runs are identical to the canonical run_N3_N40_stage123_v1.py (SHA256 1abf2353fee2e4f56f05e7a6f149fd086885136beb61ab571b48a56b09691567), with only the initial seed controlled.